ARadhana

AR-Guided Heritage & Campus Exploration

[TODO: one-line tagline]

Project Documentation

Team	**[TODO: team name]**
Members	**[TODO: member 1]** **[TODO: member 2]** **[TODO: member 3]**
Event	**[TODO: hackathon / course]**
Institution	**[TODO: institution]**
Date	July 11, 2026

[TODO: footer note, e.g. supervisor / mentor]

Abstract

[TODO: 150–250 word summary: what ARadhana is, the problem it solves, the approach (AR pathfinding + gamification), and the headline outcome.]

Keywords: **[TODO: augmented reality, geofencing, gamification, . . .]**

Contents

List of Figures

List of Tables

Chapter 1

Introduction

1.1 Background

[TODO: Context: heritage/campus exploration, why guided AR walks.]

1.2 Project Overview

[TODO: What ARadhana does in 3–4 paragraphs: AR pathfinder, checkpoints, milestones, side quests, quizzes, AR points, badges, leaderboard.]

1.3 Scope

[TODO: What is in scope for this build; what is explicitly out of scope.]

1.4 Report Organisation

[TODO: One line per chapter describing what the reader will find.]

Chapter 2

Problem Statement

[TODO: THE PROBLEM STATEMENT. Replace entirely — do not reuse the auto-generated draft. Suggested shape: (1) the situation today and who is affected, (2) the specific gap or pain point, (3) why existing approaches fall short, (4) the consequence of leaving it unsolved.]

2.1 Motivation

[TODO: Why this problem is worth solving now, and why this team.]

Chapter 3

Objectives

3.1 General Objective

[TODO: The single overarching goal.]

3.2 Specific Objectives

- **[TODO: objective 1 (e.g. verify physical presence via geofencing)]**
- **[TODO: objective 2 (e.g. guide visitors with AR waypoints)]**
- **[TODO: objective 3 (e.g. drive engagement with gamification)]**
- **[TODO: objective 4]**

Chapter 4

Market Analysis

4.1 Target Users

[TODO: Segments and their needs.]

4.2 Existing Solutions

[TODO: Competitor / alternative overview table.]

4.3 Differentiation

[TODO: What ARadhana does that the alternatives do not.]

Chapter 5

Design Thinking

5.1 Empathise

[TODO: User research / assumptions.]

5.2 Define

[TODO: Point-of-view statements.]

5.3 Ideate

[TODO: Options considered.]

5.4 Prototype & Test

[TODO: What was prototyped, what feedback changed.]

Chapter 6

System Architecture

6.1 High-Level Architecture

[TODO: Diagram + narrative: React client, Express File Cluster server, MongoDB, WebSocket pathfinder channel.]

6.2 Client

[TODO: SPA structure, routing, role-based experience (user vs admin), RTK Query.]

6.3 Server

[TODO: File-based routing, auth strategy, WebSocket proximity engine.]

6.4 Real-Time Location Pipeline

[TODO: GPS watch → WS location frames → haversine proximity → arrival confirmation → dynamic geofence threshold served from the place record.]

Chapter 7

Database Design

7.1 Entity Overview

[TODO: ER-style overview of User, VisitingPlace, VisitingRoutes, UserProgress, Quiz, QuizAttempt, ArPoints.]

7.2 Schema Reference

[TODO: One subsection per model; use lstlisting with language=json.]

7.3 Denormalisation Decisions

[TODO: Why visitor_count is a counter, why AR points are a ledger, badge permanence via ledger + profile milestones.]

Chapter 8

Gamification Engine

8.1 AR Points

[TODO: Earning table: quiz 10/20/25/30/35, side quest 25, milestone 10, place completion 20. Privacy: points are never public.]

8.2 Quizzes

[TODO: One quiz per place, 5 questions, unlock-on-completion, single attempt, answer review.]

8.3 Badges & Milestones

[TODO: Badge rules, permanence across revisits.]

8.4 Leaderboard

[TODO: Metrics (milestones / places / side quests), why AR points are excluded.]

8.5 Redemption

[TODO: Rewards catalog and the spend ledger.]

Chapter 9

AR Pathfinder

9.1 Geofenced Checkpoints

[TODO: 10 m dynamic threshold, physical-presence guarantee, confirmation flow.]

9.2 AR Guidance

[TODO: A-Frame + AR.js location-based scene, compass HUD, anchored video.]

9.3 Waypoint Types

[TODO: start / node / milestone / side_quest / end and their behaviours.]

9.4 Demo Mode

[TODO: Simulated walk for judging without GPS.]

Chapter 10

Technology Stack

10.1 Frontend

[TODO: React 19, Redux Toolkit Query, Tailwind CSS 4, lucide-react, cartoon-avatar, A-Frame/AR.js.]

10.2 Backend

[TODO: Express File Cluster, MongoDB/mongoose, ws, jose, bcrypt.]

10.3 Tooling

[TODO: Vite, TypeScript, ESLint, tsx.]

Chapter 11

Results

11.1 Delivered Features

[TODO: Checklist of shipped functionality.]

11.2 Screenshots

[TODO: Figures: explore, AR view, quiz, redeem, leaderboard, admin dashboard.]

11.3 Field Test

[TODO: Observations from the Orchid campus walk.]

Chapter 12

Limitations & Future Work

12.1 Known Limitations

[TODO: GPS accuracy vs 10 m geofence, HTTPS requirement for camera, avatar CDN dependency, . . .]

12.2 Future Work

[TODO: Roadmap items.]

Chapter 13

Conclusion

[TODO: Close the loop: restate the problem, what was built, what it proves.]

Bibliography

[1] AR.js Organization, "Ar.js — augmented reality on the web." https://ar-js-org.github.io/AR.js-Docs/. Accessed: [**TODO: date**].

[2] Supermedium, "A-frame: Web framework for building virtual reality experiences." https://aframe.io. Accessed: [**TODO: date**].

[3] R. W. Sinnott, "Virtues of the haversine," *Sky and Telescope*, vol. 68, no. 2, p. 159, 1984.

Chapter A

API Reference

[TODO: Endpoint tables grouped by area. Example row format:]

Chapter B

Folder Structure

[TODO: Annotated tree of client/ and server/.]